EXPERIMENT NO:10

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture

# Generate sample dataset
np.random.seed(42)

X1 = np.random.randn(100, 2) + [2, 2]
X2 = np.random.randn(100, 2) + [7, 7]

X = np.vstack((X1, X2))

# Create Gaussian Mixture Model
model = GaussianMixture(n_components=2, random_state=42)

# Train model using EM algorithm
model.fit(X)

# Predict clusters
labels = model.predict(X)

print("EM Algorithm Results")
print("Number of Clusters:", model.n_components)
print("Number of Iterations:", model.n_iter_)
```

```
print("Converged:", model.converged_)
print("Cluster Centers:")
print(model.means_)

# Graph 1: Cluster visualization
plt.figure(figsize=(8, 5))

plt.scatter(
    X[:, 0], X[:, 1],
    c=labels, cmap='viridis',
    s=50, edgecolors='black'
)

plt.scatter(
    model.means_[:, 0], model.means_[:, 1],
    color='red', marker='X', s=200,
    label='Cluster Centers'
)

plt.title("Expectation-Maximization Clustering")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.legend()
plt.grid(True)
plt.show()
```

```
# Graph 2: Cluster distribution
counts = np.bincount(labels)

plt.figure(figsize=(6, 4))
plt.bar(
    ['Cluster 0', 'Cluster 1'],
    counts,
    color=['skyblue', 'orange']
)

plt.title("Number of Samples in Each Cluster")
plt.xlabel("Clusters")
plt.ylabel("Number of Samples")
plt.show()
```

OUTPUT:

```
EM Algorithm Results
Number of Clusters: 2
Number of Iterations: 2
Converged: True
Cluster Centers:
[[1.88499152 2.03519083]
 [7.12924798 7.0438843 ]]
```



